

# ADITYA NAIDU

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## EDUCATION

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### ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH (AITR) *Indore, MP*

*B.Tech in Computer Science & Engineering (AI/ML), CGPA: 6.1/10 Expected Sept. 2027*

Relevant Coursework: Robotics, Control Systems, Computer Vision, Embedded Systems, [add more]

### A.P. NARMADA HR. SEC. SCHOOL (MP Board) *Jabalpur, MP*

*Higher Secondary Certificate (10+2), Science — 72.8%, First Division March 2023*

### ST. ALOYSIUS SCHOOL (CBSE) *Jabalpur, MP*

*Secondary School Certificate (10th) — 79.4% 2020*

## TECHNICAL SKILLS

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**Languages:** C++, Python, C

**Frameworks & Tools:** ROS2 (Humble), IPG CarMaker, OpenCV, ONNX Runtime (GPU), PyTorch, Flask, FastAPI, Web3.py, Ollama, Git, colcon, WSL2/Ubuntu 22.04

**Domains:** Autonomous vehicles/ADAS, sensor fusion, computer vision, applied ML, cryptography/blockchain

## PROJECTS

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### Autonomous Emergency Braking (AEB) System — Team Acroracerz *Jan 2026 – Present*

*aBAJA SAE India 2026 (Vehicle #264001)*

- Built a ROS2 package (pycarmaker) bridging IPG CarMaker simulation (Windows) with a ROS2 workspace on WSL Ubuntu 22.04, publishing GNSS pose and vehicle speed, and integrating a LiDAR RSI sensor (spherical-to-Cartesian conversion, PointCloud2 output) for obstacle detection.
- Diagnosed and resolved recurring colcon build failures across a multi-package ROS2 workspace, including package.xml export configuration and dependency conflicts.

### Real-Time Lane Detection Pipeline (TwinLiteNet) *2025 – 2026*

- Developed a real-time C++ lane detection system using TwinLiteNet (ONNX) with OpenCV and ONNX Runtime GPU for inference.
- Designed a temporal smoothing algorithm — persistent lane tracking with exponential smoothing and slope-consistency gating — to stabilize lane estimates frame to frame.
- Validated performance against both IPG CarMaker simulation footage and real-world driving video.

### Crunchy-Local — Local Anime Streaming App *2026*

- Built a self-hosted media app with FastAPI and Jinja2, integrating the Jikan v4 (MyAnimeList) API for metadata.
- Implemented automatic local library detection and folder organization logic matched against API-returned titles.

### NovAI — Secondary Disaster Prediction Model *2025*

- Built an LSTM-based multi-task model in PyTorch to predict secondary disaster occurrence, type, and intensity from primary disaster features.
- Optimized training with the Intel AI Analytics Toolkit and exported the model to ONNX for cross-platform deployment.
- Served real-time predictions through a Flask web interface.

### Cryptix-Q — Post-Quantum File Encryption Platform *2025*

- Co-developed a Flask app implementing lattice-based (post-quantum) encryption for secure file storage, benchmarked against an RSA baseline.
- Integrated Web3.py with a local Ganache blockchain to record encrypted file transactions, and built a decryptor tool to test both schemes against simulated attacks.
- Team project (Team CIPHER, 3 members); implemented authentication and the encryption/blockchain backend.

### AI Meeting Assistant *2025*

- Built a Flask app automating post-meeting workflows: real-time transcript ingestion, local-LLM analysis (Ollama/LLaMA3), and structured minutes published to Notion and Slack.
- Designed a modular multi-agent architecture (Coordinator, Virtual Assistant, Data Analysts, Watchdog) using the Palette orchestration framework.

## LEADERSHIP & ADDITIONAL EXPERIENCE

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### Team Acroracerz — Captain & AI/ADAS Developer *Feb 2026 – Present*

*aBAJA SAE India, Vehicle #264001*

- Led the migration of the team's vehicle software stack from ad hoc scripts to a distributed, ROS2-based architecture.
- Developed core autonomy modules (Lane Keep Assist, Traffic Light & Sign Detection) and an Endurance Arbitrator fusing lateral and longitudinal control commands, deployed to an NVIDIA Jetson Orin NX with sensor fusion (camera, radar, GNSS, IMU) at 50 Hz.
- Built a FastAPI-based team dashboard and automation scripts for data management and registration workflows.
- Represented the team in official communications; secured Drive-by-Wire Event Winner honors in Phase 3 of aBAJA SAE India 2025 (Global Automotive Research Centre, Chennai), contributing to an AIR 1 national ranking.